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Angular : Angular 9 vs Angular 16+

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📊 Modern vs Old Angular Mapping

```
| Modern (Angular 19+)          | Role                        | Old Angular Equi
| ----------------------------- | --------------------------- | ----------------
| **esbuild**                   | Fast TS → JS + bundling     | **TypeScript Com
| **Vite**                      | Dev server + HMR            | **Webpack Dev Se
| **Ivy**                       | Compiler + rendering engine | **View Engine**
| **Webpack** (modern optional) | Bundler                     | **Webpack (core
| **HMR (Vite-based)**          | Live updates without reload | **Webpack HMR (l
```

🧠 1. esbuild → What was used before?

❌ No esbuild in old Angular

Instead:

```
TypeScript (tsc)
   +
Webpack (ts-loader / babel-loader)
```

🔧 Old Flow

```
TS → tsc → JS → Webpack bundles → Browser
```

👉 Problems:

- 🐢 Slow rebuilds
- Heavy memory usage
- Full rebuild on every change

🚀 2. Vite → What was used before?

👉 Replaced by:

Webpack Dev Server

Old behavior:

```
Code change → Webpack rebuild → Full page reload
```

❌ Slow
❌ No instant updates

🧠 3. Ivy → What was used before?

👉 Replaced: View Engine

View Engine flow:

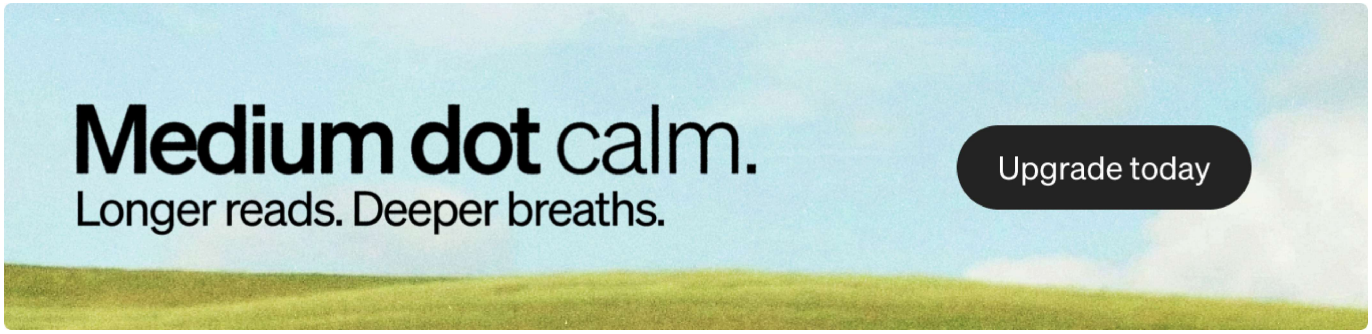
```
Template → Angular metadata → Runtime rendering
```

Problems:

- ❌ Large bundles
- ❌ Poor tree-shaking
- ❌ Complex generated code

📦 4. Webpack → Old Angular reality

👉 This one **did NOT change**



In old Angular:

> **Webpack was EVERYTHING**

Webpack handled:

- Bundling
- Dev server
- HMR
- Loaders (TS, CSS)
- Optimization

Old Architecture

```
Webpack
 ├── Bundling
 ├── Dev Server
 ├── HMR
 ├── Loaders
 └── Plugins
```

👉 Monolithic tool

🔄 5. HMR → What was used before?

👉 Webpack HMR

Behavior:

```
Code change → Rebuild chunk → Replace module
```

Problems:

- ❌ Hard to configure
- ❌ Not reliable in Angular
- ❌ Often fallback to full reload

🔥 Full Architecture Comparison

🟠 OLD Angular (v8 and below)

```
Angular App
   ↓
View Engine 🧠
   ↓
TypeScript (tsc)
   ↓
Webpack 📦
   ↓
Webpack Dev Server 🚫
   ↓
Browser (full reload)
```

🟢 NEW Angular (v19)

```
Angular App
   ↓
Ivy 🧠
   ↓
esbuild ⚡
   ↓
Vite 🚀
   ↓
Browser (HMR ⚡)
```

⚠️ Key Problems in Old Angular

❌ Slow builds

- Every change → full rebuild

❌ Poor DX (Developer Experience)

- Reloads instead of updates

❌ Large bundles

- View Engine inefficiency

❌ Tight coupling

- Everything depended on Webpack

🧠 Key Evolution Insight

> Angular moved from **monolithic (Webpack)**
> to **modular architecture (Vite + esbuild + Ivy)**

🔁 Direct Replacement Summary

```
| Old Tool              | Modern Replacement |
| --------------------- | ------------------ |
| View Engine           | Ivy                |
| tsc (build)           | esbuild            |
| Webpack Dev Server    | Vite               |
| Webpack (all-in-one)  | Split tools        |
| Webpack HMR           | Vite HMR           |
```

💡 One-Line Takeaway

> Old Angular = Webpack-centric
>
> Modern Angular = **Tool-separated architecture**

🚀 Final Insight

The biggest shift:

❌ Before:

> One tool (Webpack) doing everything

✅ Now:

Specialized tools:

- Ivy → framework logic
- esbuild → speed
- Vite → dev experience

Angular UI Coding Technology

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No responses yet

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What are your thoughts?